Interplay of Public and Private Educational Spending: Macroeconomic Implications

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Abstract

This paper examines how the composition of education financing influences economic performance and inequality. I develop an overlapping-generations model with heterogeneous agents in which human capital is accumulated through both public and private education spending. The model predicts that shifting the financing mix toward public provision fosters economic growth and reduces income inequality. These predictions are tested using cross-country dynamic panel data models. The empirical results show that a larger public-private education spending gap, where public spending exceeds private, significantly increases GDP per capita and lowers disposable income inequality. The findings underscore that not only the level but also the structure of education financing plays an important role in shaping macroeconomic and distributional outcomes.

Keywords: Human Capital · Income Inequality · GDP per capita · Education
JEL Classification: H52 · I21 · I22 · I24 · O40

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1 Introduction

In recent years, many countries have experienced rising income inequality (Alvaredo et al. 2018b) and declining intergenerational mobility (Narayan et al. 2018). These trends have renewed attention to the role of education in shaping economic opportunity and long-run growth. Because education is a key channel through which skills are transmitted across generations, the way it is financed, through public or private sources, can have important implications for both growth and income distribution. This paper examines the macroeconomic consequences of education financing, focusing specifically on the balance between public and private educational expenditures. These questions are not merely theoretical. In the Netherlands, for example, a 2015 reform replaced universal student grants with income-contingent loans, sparking debate over access and inequality (de Gendre & Kabátek 2021). In Germany, tuition fees were abolished across all federal states by 2014 after a period of reintroductions (Minor 2023). Such policy shifts illustrate the importance of understanding how the public–private mix of education finance affects both growth and distribution.

A key motivation for studying the composition of education spending is that it shapes access to human capital in ways that total expenditure alone does not capture. When education systems rely heavily on private spending, households face higher out-of-pocket costs, which may discourage investment among students from low-income backgrounds, particularly where credit markets are imperfect (Chetty et al. 2014). By contrast, a higher public share reduces financial barriers and broadens access to skills. OECD evidence suggests that systems with stronger public cost-sharing tend to exhibit higher intergenerational mobility, smaller skill gaps, and better learning outcomes among disadvantaged groups (OECD 2018).

The structure of education finance also interacts with long-run growth. In highly privatized systems, wealthier families can invest more in human capital while poorer households underinvest, widening skill gaps and slowing aggregate growth. Public financing can mitigate these disparities by lowering the effective cost of educational investment and expanding participation in skill formation, while private provision may contribute to quality improvements through competition and innovation (Hanushek & Woessmann 2011). These considerations highlight why the balance between public and private spending, rather than either component alone, is central to understanding the evolution of growth and inequality.

Human capital investments are widely recognized as a key driver of both growth and equality.[1] The preference for public or private education varies across countries and has been studied extensively. Glomm & Ravikumar (2003) and Dotti (2019) show that public education can reduce inequality under certain conditions, while De la Croix & Doepke (2009) argue that in unequal societies wealthier households may opt for private education, indirectly affecting public school quality. Interestingly, when human capital is initially more equally distributed, systems relying more heavily on private education can sometimes generate higher growth (De la Croix & Doepke 2004). Yet much of this literature examines public and private education in isolation, leaving open the question of how changes in their relative importance jointly affect growth and inequality. This paper therefore asks three central questions: what is the optimal

[1] For recent reviews of returns to human capital investment see Heckman et al. (2018), Flabbi & Gatti (2018), and Rossi (2020). For links between education and economic growth see Ogbeifun & Shobande (2021).

balance between public and private education spending? How does shifting the mix toward public financing affect GDP per capita and income inequality? And should governments expand public spending relative to private contributions?

To address these questions, the paper combines a theoretical framework with an empirical analysis. The theoretical contribution is an overlapping-generations (OLG) model in which growth is driven by human capital accumulated through both publicly and parentally financed education. The theoretical model captures the broader mechanisms of human capital accumulation, while the empirical analysis focuses specifically on the financing of tertiary education, where public and private spending can be measured most consistently across OECD countries. In contrast to earlier contributions such as De la Croix (2001), the framework emphasizes the policy trade-off between public and private education financing rather than focusing primarily on credit frictions.

The model shows how the composition of education finance shapes the joint evolution of growth, inequality, and welfare. To ensure that inequality remains meaningful over time, the framework incorporates heterogeneous permanent learning abilities, generating persistent differences in human capital accumulation across households and a non-degenerate stationary distribution of human capital, as in heterogeneous-agent human capital models such as Artige & Cavenaile (2023). Within this framework, greater reliance on public education spending tends to promote growth and reduce inequality by lowering the effective cost of educational investment for households with lower human capital. Among the simulated policy counterfactuals, reforms combining progressive taxation with higher public education spending generate the largest aggregate welfare gains.

The theoretical analysis is complemented by an empirical application using cross-country panel data. Dynamic panel methods are used to address persistence and potential endogeneity, while the limitations of available data on private education spending are discussed. The empirical results broadly support the theoretical predictions: countries with higher public spending relative to private spending tend to exhibit higher GDP growth and lower inequality, although the marginal effects diminish at very high levels of public provision. Together, these findings suggest that the composition of education spending matters as much as its total level, with public financing playing a crucial role in broadening access to skills and supporting inclusive growth, while private spending remains valuable when it complements rather than substitutes for adequate public provision. The remainder of the paper is organized as follows. Section 2 presents the theoretical model. Section 3 describes the empirical analysis. Section 4 discusses the results and policy implications. Section 5 concludes.

2 Theoretical Model

The model examines the policy trade-off between public and private education financing and how their balance affects economic growth and income inequality. Human capital formation relies on both public inputs, such as publicly provided schooling, and private household investments, including tutoring, health, extracurricular activities, and educational technology. The framework follows the OLG structure of De la Croix (2001) but shifts the emphasis from credit frictions to the composition of education financing and its implications for human capital accumulation and income distribution.

A key feature is that education preferences are non-homothetic: the importance households place on investing in their children's education depends on their level of human capital, capturing the empirical observation that higher-income households allocate a larger share of resources to education-related activities. Human capital is produced using both publicly funded and parentally financed inputs, with intergenerational transmission reinforcing these effects. To prevent the degenerate long-run distribution implied by a fully deterministic concave technology, the model incorporates heterogeneity in learning abilities, generating persistent inequality and a stationary income distribution that can be mapped to empirical measures such as the Gini coefficient.

2.1 Environment

Consider an overlapping-generations economy in which individuals live for three periods and die at the end of the third. Each individual has one parent and one child, so the population size is constant and normalized to one. Individuals share identical preferences but differ in their ability to transform educational inputs into human capital. Differences across households therefore arise from heterogeneity in inherited human capital and permanent ability draws, which translate into differences in income.

In the first period of life, individuals do not work or consume; they only accumulate human capital. In the second period, they supply one unit of labor inelastically, earn wages, consume, and save for retirement. In the third period, they retire and consume their accumulated savings. At $t = 0$, the initial generation is endowed with human capital h_{j0} and a permanent ability draw a_j. Ability follows a log-normal distribution with mean one and scales the productivity of educational investments without changing over the life cycle. To focus on the interaction between public and private education financing, the model abstracts from credit markets.

2.1.1 Household Optimization

Agents born in period $t-1$ (generation t) have preferences over three components: consumption in adulthood c_t, consumption in retirement d_{t+1}, and investment in their child's education e_t. Lifetime utility is given by

$$\max_{c_t, d_{t+1}, e_t} \quad u(c_t, d_{t+1}, e_t) = \ln(c_t) + \beta \ln(d_{t+1}) + \gamma_i \ln(e_t), \tag{1}$$

where β is the discount factor for retirement consumption.

The parameter γ_i captures heterogeneity in education preferences and introduces non-homotheticity into the model:

$$\gamma_i = \frac{\gamma_0}{1 + \phi_{nh} \frac{h_i}{\bar{h}_0}}, \tag{2}$$

where $\gamma_0 > 0$ is the baseline preference for education and ϕ_{nh} governs the strength of the non-homotheticity mechanism. This specification implies that the marginal utility weight placed on education varies with relative human capital, generating systematic differences in private education spending across households.

Labor income is taxed at a proportional rate τ_t, which applies to earnings from supplying efficient labor units. Adults earn after-tax labor income $(1-\tau_t)w_t h_t$ and allocate it across current consumption, educational investment, and retirement consumption. The household problem can therefore be written using a present-value budget constraint:

$$c_t + e_t + \frac{d_{t+1}}{R_{t+1}} = (1-\tau_t)w_t h_t, \tag{3}$$

where $R_{t+1} = 1 + r_{t+1}$ is the gross return on savings.

Equivalently, households save an amount s_t in adulthood such that retirement consumption satisfies

$$d_{t+1} = R_{t+1} s_t. \tag{4}$$

2.1.2 Human Capital Accumulation

Human capital is accumulated during the first stage of life through both public and private educational inputs. The human capital production function is

$$h_{i,t+1} = a_i \Psi g_t^{\lambda} e_{i,t}^{\theta} h_{i,t}^{1-\lambda-\theta}, \tag{5}$$

where $\Psi > 0$ is a productivity parameter and a_i denotes the individual ability draw.

Ability is drawn from a stationary distribution and captures heterogeneity in innate learning capacity across households. The parameter a_i scales the productivity of both public and private educational inputs, implying that individuals with higher ability convert educational investments into human capital more efficiently. Ability is permanent over the life cycle and therefore generates persistent differences in human capital accumulation across households.

Public education spending g_t is provided uniformly by the government, while $e_{i,t}$ represents parental spending on children's education. The parameters satisfy

$$0 < \lambda < 1, \quad 0 < \theta < 1, \quad \lambda + \theta < 1.$$

The parameter λ captures the elasticity of human capital with respect to public education spending, while θ captures the elasticity with respect to private parental investment. Public education therefore affects all households symmetrically, whereas private education decisions generate heterogeneity in educational investment across families.

The term $h_{i,t}^{1-\lambda-\theta}$ introduces partial intergenerational persistence of human capital, generating dynamic inequality even in the absence of borrowing constraints.

2.1.3 Firms

Production is carried out by competitive firms using physical capital K_t and human capital H_t according to the Cobb–Douglas technology

$$Y_t = K_t^{\alpha} H_t^{1-\alpha}. \tag{6}$$

Human capital is the aggregate of individual efficiency units:

$$H_t = \int h_{i,t}\, d\Gamma_t(h). \tag{7}$$

Competitive factor markets imply that labor and capital are paid their marginal products:

$$w_t = (1-\alpha)\frac{Y_t}{H_t}, \qquad r_t = \alpha\frac{Y_t}{K_t}. \tag{8}$$

2.1.4 Government

An infinitely lived government finances public education through a proportional tax on labor income. Total tax revenue is

$$T_t = \tau_t w_t H_t. \tag{9}$$

A fixed share $\phi_g \in (0,1)$ of this revenue is allocated to public education spending:

$$g_t = \phi_g \tau_t w_t H_t. \tag{10}$$

The remaining share of tax revenue finances other government expenditures that do not directly enter the human capital production function:

$$G_t = (1-\phi_g)T_t. \tag{11}$$

These expenditures represent public consumption and other government programs treated as exogenous from the perspective of household decisions.

2.2 Equilibrium

A competitive equilibrium consists of sequences

$$\{c_{i,t}, d_{i,t+1}, e_{i,t}, s_{i,t}, h_{i,t}\}, \{w_t, r_t\}, \{g_t\},$$

and distributions Γ_t such that:

1. Households maximize utility subject to their budget constraints, taking prices and government policy as given.
2. Firms maximize profits and satisfy the factor pricing conditions.
3. The government budget constraint holds.
4. Capital and labor markets clear.
5. The distribution of human capital evolves according to the accumulation technology
$$h_{i,t+1} = a_i \Psi g_t^{\lambda} e_{i,t}^{\theta} h_{i,t}^{1-\lambda-\theta},$$
which induces the law of motion for the distribution Γ_t.

Aggregate capital evolves according to household savings:

$$K_{t+1} = \int s_{i,t}\, d\Gamma_t. \tag{12}$$

Because public education spending depends on aggregate tax revenue, fiscal policy feeds back into the human capital accumulation process and therefore affects both growth and the evolution of inequality over time.

2.3 Numerical Simulation

The numerical simulation quantifies the mechanisms of the model and evaluates policy counterfactuals affecting taxation and education finance. Because the model features heterogeneous households and non-homothetic education preferences, closed-form aggregate dynamics are generally not available, so simulation allows the full distribution of human capital to evolve endogenously across generations.

The structural parameters are calibrated to match empirical counterparts for OECD economies around 2017, with targets summarized in Table 1. Public education expenditure is matched by choosing ϕ_g, the share of government revenue allocated to education, which together with the tax rate τ and labor income share $(1-\alpha)$ determines the equilibrium ratio of public education spending to GDP,

$$\frac{g_t}{Y_t} = \phi_g \tau (1-\alpha). \tag{13}$$

Private education spending is disciplined by the preference parameter γ_0, which governs parental investment in children's education. On the household side, the discount factor is set to $\beta = 0.8$ (Galor & Weil 2000), the baseline education preference parameter to $\gamma_0 = 0.08$, and the non-homotheticity parameter to $\phi_{nh} = 0.28$. The human capital production elasticities are $\lambda = 0.25$ and $\theta = 0.25$ (Glomm & Ravikumar 1992). The three parameters γ_0, ϕ_{nh}, and $\sigma_a = 0.28$ are jointly calibrated to match private education spending as a share of GDP (0.73%), the 90/10 income ratio (5.0), and the bottom-50% income share (0.25) (World Inequality Database 2022). On the production side, the capital share is $\alpha = 0.30$ (King & Rebelo 1999), the tax parameter $\tau = 0.34$ (OECD 2023), and the government allocates $\phi_g = 0.198$ of revenue to education, reproducing the observed public education spending ratio.

Table 1
Calibration Targets

	Model	Data
Public Education (% of GDP)	4.71	4.71
Private Education (% of GDP)	0.78	0.73
Saving Rate (% of GDP)	28.35	22.00
Gini Coefficient	0.33	0.30
90/10 Income Ratio	5.22	5.00
Bottom 50% Income Share	0.25	0.25

The calibration reproduces key empirical patterns of OECD economies. Public education spending matches its empirical counterpart by construction, while private education spending and inequality measures are close to their benchmarks. The simulated saving rate is somewhat higher than in the data, reflecting the model's abstraction from government transfers, pension systems, and capital depreciation. Given

these parameters, the model is simulated forward across generations until the distribution of human capital converges to a stationary equilibrium, after which policy experiments are conducted by perturbing fiscal parameters and recomputing transition dynamics and steady-state outcomes.

Figure A.1.1 summarizes the baseline transition dynamics. Panel (a) shows the evolution of the Gini coefficient as the economy converges toward its steady-state equilibrium, while Panel (b) reports the corresponding shares of public and private education spending as a percentage of GDP. Public spending is distributed uniformly across households while private spending depends on household income and human capital, so households with higher human capital invest more in private education for their children. Through this interaction, the public–private composition of education finance shapes both the distribution of skills and aggregate economic outcomes over time.

2.3.1 Policy Counterfactuals

To quantify the mechanisms described above, I conduct a set of policy counterfactuals using the calibrated model. The objective of these simulations is not to generate precise forecasts but to evaluate how alternative fiscal policies affect the long-run distribution of income, aggregate output, and welfare. Welfare effects are measured using consumption-equivalent variation (CEV), which allows the model to assess how different groups gain or lose relative to the baseline steady state.

Table 2

Policy Counterfactuals: Baseline vs. Alternative Policies

	Baseline	CF1	CF2	CF3	CF4
Panel A: Policy variables (levels)					
Public Education (% GDP)	4.71	4.76	4.71	4.76	4.66
Private Education (% GDP)	0.78	0.78	0.81	0.81	0.80
Eff. Tax Rate: Top 10% (%)	34.00	34.00	39.58	39.57	34.00
Eff. Tax Rate: Bottom 50% (%)	34.00	34.00	25.57	25.60	34.00
Panel B: Economic outcomes (percent change relative to baseline)					
Aggregate Welfare (CEV %)	–	+2.12	+10.12	+12.43	-2.11
GDP (short-run change)	–	+0.17%	-0.09%	+0.08%	-0.18%
Gini Coefficient	0.34	-0.25%	-8.98%	-9.20%	+0.25%
Top 10% Income Share	0.23	-0.20%	-6.32%	-6.48%	+0.20%
Bottom 50% Income Share	0.26	+0.23%	+8.39%	+8.59%	-0.23%

Notes: Panel A reports levels of policy variables under the baseline and each counterfactual scenario. Panel B reports percentage changes in economic outcomes relative to the baseline steady state. Welfare changes are measured as consumption-equivalent variation (CEV). Source: Author's calculations.

Four counterfactual experiments are considered. The first increases public education spending relative to the baseline while leaving the tax schedule unchanged. The second introduces progressive labor-income taxation while maintaining overall government revenue constant. The third combines both reforms, while the fourth reduces public education spending, representing a shift toward greater reliance on private education financing. Table 2 summarizes the steady-state outcomes of these experiments relative to the baseline equilibrium.

Figures 1 and 2 illustrate the transition dynamics of inequality and output under the alternative policy scenarios. The first counterfactual isolates the education-spending channel. Increasing public education spending raises the public education share of GDP from 4.71 percent to 4.76 percent. Private education spending declines slightly as households partially substitute public provision for private investment. The resulting reduction in inequality is modest: the Gini coefficient falls by approximately 0.25 percent relative to the baseline. At the same time, output increases slightly as higher public spending strengthens human capital accumulation. Welfare gains are positive but relatively small, with aggregate welfare increasing by roughly 2 percent in CEV terms.

The second counterfactual introduces progressive labor-income taxation while keeping total tax revenue constant. Following Heathcote et al. (2017) (HSV), after-tax labor income follows

$$\ell^a_{i,t} = \lambda^{\text{HSV}}_t \ell^{1-\tau_p}_{i,t}, \tag{14}$$

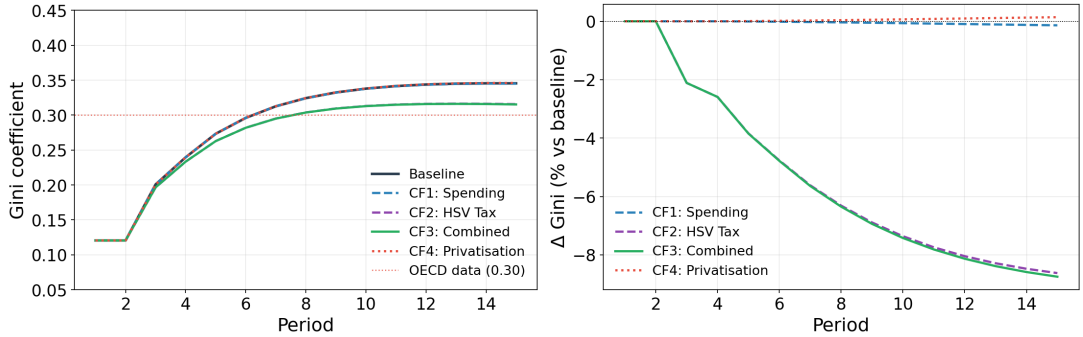
where $\ell_{i,t} = w_t h_{i,t}$ denotes pre-tax labor income, τ_p governs the degree of tax progressivity, and λ^{HSV}_t is a scaling parameter chosen each period to keep total tax revenue equal to the baseline level $T_t = \tau w_t H_t$, distinct from the public education elasticity λ in equation 5. In counterfactuals CF2 and CF3, the flat-tax labor income term $(1-\tau_t)w_t h_{i,t}$ in the household budget constraint (equation 3) is replaced by the progressive after-tax income $\ell^a_{i,t}$ defined above. In this counterfactual, the progressivity parameter is set to $\tau_p = 0.15$.

This reform increases effective tax rates for higher-income households while reducing them for lower-income households. In particular, the effective tax rate faced by the top decile rises from 34 percent to approximately 39.6 percent, while the rate for the bottom half declines to about 25.6 percent. Because the reform is revenue neutral, total public education spending remains unchanged. The redistribution of disposable income generates a substantial decline in inequality, with the Gini coefficient falling by roughly 9 percent relative to the baseline. Output declines slightly as higher tax progressivity weakens incentives for saving and investment. From a welfare perspective, however, the reform generates large gains for lower-income households, while higher-income households experience welfare losses.

The third counterfactual combines progressive taxation with higher public education spending. This reform produces the largest reduction in inequality, with the Gini coefficient declining by about 9.2 percent relative to the baseline. The additional public education spending strengthens human capital accumulation and largely offsets the negative output effects of progressive taxation, leaving GDP close to its baseline level. The combined reform also produces the largest welfare gains in the model, with aggregate welfare increasing by more than 12 percent in CEV terms.

The final counterfactual examines the effects of reducing public education spending. Public education expenditure falls to 4.66 percent of GDP, increasing reliance on private educational investment. As a result, inequality rises slightly relative to the baseline and output declines modestly. Welfare also falls across the income distribution, with aggregate welfare declining by about 2 percent in CEV terms.

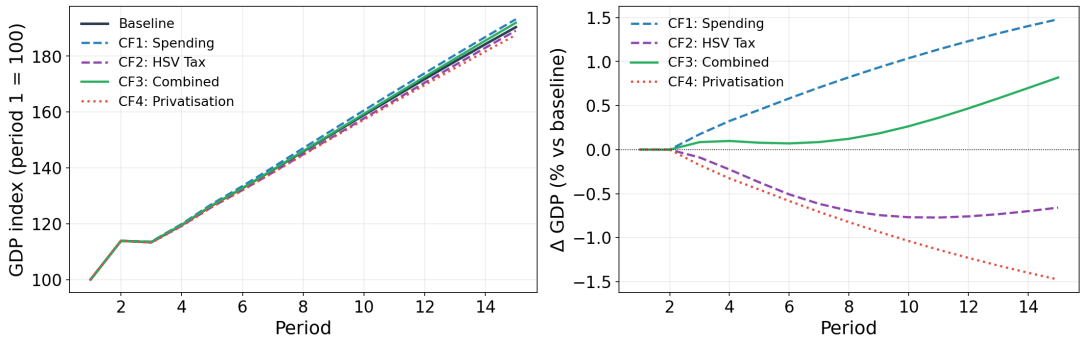
Figure 3 summarizes the welfare implications of these policy experiments. Panel (a) reports welfare changes for key income groups, while Panel (b) illustrates how welfare gains and losses vary across the full income distribution. The results highlight the strongly progressive nature of the tax reform, with welfare gains concentrated among



(a) Across policy scenarios (b) Change relative to baseline

Figure 1: Inequality dynamics across model scenarios

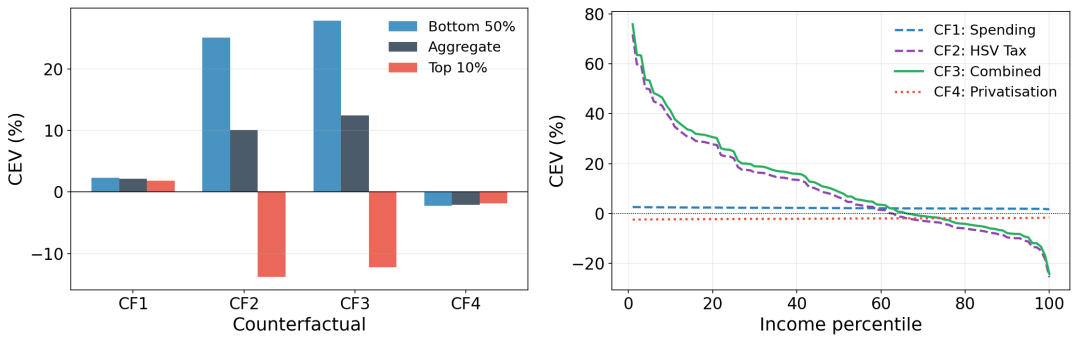
Notes: Panel (a) shows the evolution of the Gini coefficient under alternative model scenarios: higher public education spending (CF1), progressive taxation (CF2), a combined policy (CF3), and reduced public education spending (CF4). Panel (b) reports the percentage deviation of the Gini coefficient relative to the baseline equilibrium. Source: Author's calculations.



(a) Across policy scenarios (b) GDP dynamics across model scenarios

Figure 2: GDP dynamics across model scenarios

Notes: Panel (a) shows the evolution of GDP under alternative model scenarios: higher public education spending (CF1), progressive taxation (CF2), a combined policy (CF3), and reduced public education spending (CF4). Panel (b) reports the percentage deviation of GDP relative to the baseline equilibrium for each scenario. Source: Author's calculations.



(a) Across policy scenarios (b) Across income percentiles

Figure 3: Welfare effects of policy counterfactuals

Notes: Panel (a) reports consumption-equivalent variation (CEV) for different income groups under the policy counterfactuals: increased public education spending (CF1), progressive taxation (CF2), the combined reform (CF3), and reduced public education spending (CF4). Panel (b) shows the distribution of welfare effects across the income distribution, where the horizontal axis represents income percentiles and the vertical axis reports the CEV relative to the baseline steady state. Positive values indicate welfare gains, while negative values indicate welfare losses. Source: Author's calculations.

lower-income households and welfare losses among higher-income households. The magnitude of these distributional effects is substantial. Under the progressive tax reform (CF2), households in the bottom half of the income distribution experience welfare gains of roughly 25 percent in consumption-equivalent terms, while the top decile experiences a welfare loss of about 13.8 percent. The combined policy reform (CF3) further amplifies these gains, with bottom 50% households gaining nearly 28 percent in CEV terms. Detailed welfare results are reported in Table A.3.1 in Appendix A.3.

The counterfactual experiments illustrate two distinct channels through which fiscal policy affects economic outcomes. Changes in the composition of education finance influence inequality primarily through human capital accumulation, while tax progressivity affects inequality directly through redistribution of disposable income. Policies combining both mechanisms reduce inequality while maintaining strong economic performance, as the output gains from broader human capital formation largely offset the investment disincentives of progressive taxation.

The model delivers three predictions that guide the empirical analysis. A larger public share of education financing raises long-run GDP per capita by lowering the effective cost of educational investment and expanding the aggregate stock of skills. Heavier reliance on private education spending amplifies inequality because higher-income households invest more in their children's education. And these effects should be detectable in cross-country data as systematic differences in growth and inequality associated with differences in the composition of education finance. Section 3 examines whether the data support these predictions.

3 Empirical Analysis

The analysis examines the relationship between the public–private composition of education spending and two outcomes: disposable income inequality and GDP per capita. To address the persistence and potential endogeneity inherent in these relationships, the empirical strategy relies on the System GMM estimator of Arellano & Bover (1995) and Blundell & Bond (1998), which combines equations in differences and levels and instruments endogenous variables using their own lagged differences and levels. Instrument validity is assessed using the Hansen test for overidentifying restrictions and the AR(2) test for second-order serial correlation in the differenced residuals. To avoid overfitting, the number of instruments is kept below the number of countries in each specification. All specifications are estimated with one to two lag periods, country fixed effects to account for the substantial cross-country heterogeneity documented in Figures B.0.5 and B.0.6, and Windmeijer-corrected standard errors.

3.1 Empirical Framework

3.1.1 Income Inequality

The first dependent variable is the Gini coefficient of disposable income. The baseline specification is:

$$\log \mathrm{GINI}_{it} = \beta\, \mathrm{PP}_{it} + \alpha_1 \log \mathrm{GINI}_{i,t-1} + \alpha_2 \log \mathrm{GDP}_{it} + \alpha_3 X_{it}^{gini} + \gamma_i + \epsilon_{it}, \qquad (15)$$

where PP_{it} is the public–private tertiary education spending gap, $\log \text{GINI}_{i,t-1}$ captures inequality persistence, $\log \text{GDP}_{it}$ controls for income levels, X_{it}^{gini} includes government consumption, employment in agriculture, adolescent fertility, years of compulsory education, and governance, γ_i denotes country fixed effects, and ϵ_{it} is the error term. The coefficient on PP_{it} captures the marginal effect of shifting the public–private balance rather than assuming equal productivity of both inputs.

The controls in X_{it}^{gini} follow the literature on the determinants of inequality. Government consumption affects income distribution through public sector compensation (De Mello & Tiongson 2006; Anderson et al. 2017). Employment in agriculture proxies sector dualism, since the shift toward non-agricultural employment generates between-sector inequality (Alderson & Nielsen 2002). Adolescent fertility affects the human capital accumulation of young individuals and their labor market entry (Kearney & Levine 2012). Governance captures the unequal political influence of high-income households (Chong & Gradstein 2007), while employment conditions for unskilled workers directly affect the low-income share of the distribution (Glyn 1995).

3.1.2 GDP per capita

The second dependent variable is GDP per capita. The baseline specification follows:

$$\log \text{GDP}_{it} = \Psi \, \text{PP}_{it} + \theta_1 \log \text{GDP}_{i,t-1} + \theta_2 \log \text{GINI}_{it} + \theta_3 X_{it}^{gdp} + \gamma_i + \upsilon_{it} \tag{16}$$

where PP_{it} is the public–private education spending gap, $\log \text{GDP}_{i,t-1}$ captures output persistence, $\log \text{GINI}_{it}$ controls for inequality, and X_{it}^{gdp} includes tertiary enrollment, governance, FDI, gross fixed capital formation, trade, inflation, employment in agriculture, employment rate, and government education expenditure.

The controls follow the standard growth literature. Tertiary enrollment proxies human capital, which facilitates technology adoption and innovation (Benhabib & Spiegel 1994). Governance supports political stability and growth (Islam & McGillivray 2020). FDI (Li & Liu 2005), gross fixed capital formation (Meyer & Sanusi 2019), agriculture (Tiffin & Irz 2006), and employment (Eriksson 1997) are standard engines of output growth. Trade openness (Dollar & Kraay 2003) and inflation (Barro et al. 1996) are included to control for external and monetary conditions respectively. In addition to the education spending gap, GDP per capita and income inequality are treated as endogenous and instrumented using their own lags within the System GMM framework.

3.2 Data and Descriptive Evidence

The analysis uses a balanced panel of 37 OECD countries spanning 2008 to 2019.[2] Income inequality is measured by the disposable income Gini coefficient and GDP per capita is sourced from the World Bank. Private and public expenditures on tertiary education are drawn from the OECD Education Statistics database. The main independent variable PP_{it} represents the difference between public and private spending on tertiary education as a share of GDP, with positive values indicating that public spending exceeds private.[3] Summary statistics and variable definitions are provided

[2]The list of countries is detailed in Table B.0.2 in the Appendix.
[3]A cross-country overview is provided in Figure B.0.4 in the Appendix.

in Tables B.0.3 and B.0.4 in the Appendix. Figures B.0.5 and B.0.6 document substantial cross-country heterogeneity in inequality and GDP per capita, which motivates the use of country fixed effects throughout the analysis.

3.3 Empirical Results

3.3.1 Income Inequality

Table 3
Dynamic Regression: Income Inequality

	Log Disposable Income Gini	
	(1)	(2)
L.Log D. Income Gini	0.921***	
	(0.056)	
L2.Log D. Income Gini		0.703***
		(0.131)
Public–Private Gap	−0.064***	−0.131***
	(0.021)	(0.039)
Observations	226	202
Hansen p-value	0.291	0.171
AR(2) p-value	0.887	0.064
No. Instruments	20	22
No. Countries	23	25

Note: The table reports the effects of education investment on the Log Disposable Income Gini. Standard errors are Windmeijer-corrected and robust to heteroscedasticity. Constant, country effects, and control variables are included but not reported for brevity. Significance levels are: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 3 reports the baseline dynamic specification in column (1), estimated following equation (15). The dependent variable is the log of the disposable income Gini. Consistent with the persistence typically observed in inequality dynamics, the coefficient on the lagged dependent variable is large and highly significant (0.92). The public–private education expenditure differential is negative and statistically significant, indicating that a widening gap, reflecting higher public relative to private education investment, is associated with lower income inequality. A one percentage point increase in the public–private spending gap (as a share of GDP) is associated with a 6.4 percent reduction in the disposable income Gini.

All regressions treat the disposable income Gini, GDP per capita, and the public–private expenditure differential as endogenous. Their lags serve as internal instruments, while all remaining controls enter as external instruments. The full list of instruments is provided in Section B of the Appendix. The specification passes standard diagnostic tests: the Hansen test does not reject instrument validity, and the AR(2) test fails to detect second-order serial correlation.

Column (2) presents a specification that replaces the first lag of the Gini with its second lag, allowing for a slower adjustment process. Including a second lag of inequality is an important robustness check because inequality is a slow-moving variable and its persistence can absorb substantial variation in dynamic GMM models. The equalizing effect of public–private education expenditure remains negative and significant under this alternative dynamic structure, demonstrating that our main

finding does not depend on the specific lag order and is robust to slower adjustment processes. Countries with relatively higher public education spending experience lower levels of income inequality. The two dynamic specifications demonstrate that the main result, the equalizing effect of higher public relative to private education investment, is robust to alternative assumptions about the persistence of income inequality and the structure of the dynamic model.

Table 4

Dynamic Regression: Income Distribution Shares

	Shares of Income Distribution		
	Top 1%	Top 10%	Bottom 50%
L.Log Top 1%	0.920***		
	(0.043)		
L.Log Top 10%		0.968***	
		(0.032)	
L.Log Bottom 50%			0.945***
			(0.036)
Public-Private Gap	−0.023*	−0.033*	0.024*
	(0.013)	(0.019)	(0.013)
Observations	226	217	215
Hansen p-value	0.346	0.854	0.258
AR(2) p-value	0.183	0.414	0.443
No. Instruments	26	24	24
No. Countries	29	25	24

Note: The table reports the effects of education investment on Column (1), the log of the top 1%, (2) the log of the top 10%, and (3) the bottom 50% of the income distribution. Standard errors are Windmeijer-corrected and robust to heteroscedasticity. Constant, country effects, and control variables are included but not reported for brevity. Significance levels are: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 4 presents an alternative set of dependent variables, income shares for the top 1%, top 10%, and bottom 50%, to assess whether the baseline findings are robust across different dimensions of the income distribution. These measures, drawn from the World Inequality Database Alvaredo et al. (2018a), provide a complementary view of inequality by focusing directly on distributional shares rather than a summary index. The specification follows the same dynamic structure as the baseline model, with appropriately adjusted controls.

Across all three columns, the lagged dependent variable remains large and highly persistent, as expected for distributional shares. More importantly, the coefficient on the public–private education expenditure gap retains its sign and significance in each specification. For the top 1% and top 10%, a larger public–private expenditure differential is associated with reductions of 2.3% and 3.3% in their income shares, respectively. These results indicate that greater public investment relative to private investment tends to compress the upper end of the distribution.

For the bottom 50%, the effect is reversed: the same differential increases their income share by 2.4%. This pattern is consistent with the mechanism proposed in the main analysis: public educational spending disproportionately raises the human capital of lower-income groups, who rely primarily on wage income.

Empirical evidence shows that the bottom half of the income distribution derives the vast majority of its income from wages, whereas top income groups rely more heavily on capital, business income, and financial returns Piketty et al. (2018). This

structural asymmetry helps explain why educational investment generates stronger distributional effects toward the lower tail. Taken together, these results corroborate the baseline findings: the public–private education expenditure gap reduces inequality not only through a summary measure like the Gini coefficient, but also through a direct reallocation of income shares from the top quantiles to the bottom half of the distribution.

3.3.2 GDP per capita

The results of the dynamic output specification in equation (16) and the corresponding robustness checks are presented in Table 5. As with the inequality regressions, GDP per capita, the education spending gap, and the remaining endogenous covariates are instrumented using their own lags within the System GMM framework, while additional variables serve as external instruments. A detailed list of all instruments used across specifications is provided in Section B of the Appendix.

Table 5
Dynamic Regression: Log GDP per capita

	Log GDP per capita	
	(1)	(2)
L.Log GDP p.c.	0.990***	
	(0.013)	
L2.Log GDP p.c.		0.972***
		(0.017)
Public-Private Gap	0.025**	0.030*
	(0.011)	(0.016)
Observations	233	221
Hansen p-value	0.077	0.054
AR(2) p-value	0.155	0.043
No. Instruments	27	27
No. Countries	32	32

Note: The table reports the effects of education investment on the Log GDP per capita. Standard errors are Windmeijer-corrected and robust to heteroscedasticity. Constant, country effects, and control variables are included but not reported for brevity. Significance levels are: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Column (1) reports the baseline specification using the first lag of log GDP per capita. The coefficient on the lagged dependent variable is very close to unity (0.99), reflecting the high persistence typical of macroeconomic output series. The main variable of interest, i.e., the gap between public and private education expenditure, remains positive and statistically significant, with an estimated elasticity of 2.5 percent. This confirms the core finding that a greater relative commitment to public education is associated with higher levels of economic output.

Column (2) extends the dynamic structure by replacing the first lag with the second lag of log GDP per capita. The persistence remains extremely strong (0.97), consistent with slow-moving convergence dynamics. The coefficient on the public–private spending gap remains positive and significant, with an elasticity of roughly 3 percent, slightly larger than in the baseline model. The stability of this estimate across alternative lag structures indicates that the relationship between education expenditure

composition and GDP per capita is not driven by short-term fluctuations in output dynamics.

Table 6
Dynamic Regression: Log GDP per hour

	GDP pc	GDP ph	GDP ph
L.Log GDP p.c.	0.990***		
	(0.013)		
L.Log GDP p.h.		0.967***	
		(0.029)	
L2.Log GDP p.h.			0.996***
			(0.060)
Public-Private Gap	0.025**	0.040*	0.065*
	(0.011)	(0.020)	(0.038)
Observations	233	246	226
Hansen p-value	0.077	0.102	0.285
AR(2) p-value	0.155	0.353	0.132
No. Instruments	27	26	26
No. Countries	32	35	35

Note: The table reports the effects of education investment on the log of GDP per hour. Standard errors are Windmeijer-corrected and robust to heteroscedasticity. Constant, country effects, and additional controls are included but not reported for brevity. Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 6 assesses whether the main findings for GDP per capita extend to an alternative measure of economic performance: GDP per hour worked, a standard indicator of labor productivity. Given the known strong correlation between GDP per capita and GDP per hour, this provides a meaningful validation of the baseline results. Column (1) reproduces the GDP-per-capita specification for reference. Columns (2) and (3) replace the dependent variable with the log of GDP per hour, first using a specification with one lag and then with two lags to capture slower productivity adjustment. Across both models, the lagged dependent variable remains highly persistent, with coefficients close to unity, consistent with the slow-moving dynamics of productivity.

Crucially, the coefficient on the public-private education expenditure gap remains positive and statistically significant in all specifications. The estimated magnitudes range from 4.0 percent in column (2) to 6.5 percent in column (3), slightly larger than the corresponding effect on GDP per capita (2.5 percent). This suggests that the relative strength of public education investment may influence productivity even more directly than overall economic output. Overall, the GDP-per-hour results reinforce the main conclusion: countries allocating relatively more resources to public education tend to experience stronger economic performance, whether measured through income per person or labor productivity.

While the dynamic panel approach and the use of internal instruments mitigate concerns related to endogeneity and reverse causality, the empirical results should be interpreted primarily as conditional associations rather than definitive causal estimates. Differences in education financing structures may themselves reflect deeper institutional, political, or economic characteristics across countries. The analysis therefore identifies systematic correlations consistent with the theoretical mechanism rather than establishing a fully causal effect of education spending composition.

3.4 Additional Empirical Analysis: Robustness

Tables C.0.7 and C.0.8 evaluate whether the baseline results are sensitive to alternative functional forms of the public–private education expenditure measure. Columns (1) and (2) introduce a squared term to test for nonlinearities. For GDP per capita, the quadratic term is small and insignificant, indicating no detectable curvature. For inequality, the squared term is positive and significant, suggesting diminishing marginal reductions as the gap becomes very large, though the linear term retains its expected sign and significance throughout. Column (3) incorporates total education spending to test whether baseline results are driven by overall expenditure levels rather than composition. Total spending is insignificant in both tables while the public–private gap remains strongly significant and increases in magnitude, confirming that composition rather than scale drives the observed effects. Column (4) replaces the gap with the ratio of public to private spending as an alternative measure. The ratio is negative and significant for inequality and positive and significant for GDP per capita, providing functional-form validation that the relationship holds regardless of how composition is expressed.

Tables C.0.9 and C.0.10 examine whether baseline results hold when accounting for intergenerational mobility and initial country conditions. Columns (1) and (2) introduce measures of absolute and relative intergenerational mobility, both of which are statistically insignificant, while the public–private gap remains stable in sign and significance across both outcome variables. Columns (3) and (4) incorporate interactions with initial income inequality or initial GDP levels. Interaction terms are small and insignificant throughout, and in the inequality regressions the gap remains negative and significant while in the GDP regressions it remains positive and significant, with the magnitude in Column (3) increasing relative to baseline, suggesting that accounting for initial conditions sharpens rather than weakens the underlying relationship.

Tables C.0.5 and C.0.6 assess stability across three identification challenges: alternative dynamic structures, regional and income-group heterogeneity, and reverse causality. Column (1) introduces deeper lags in the GMM instrument set, and the public–private gap retains its expected sign, significance, and magnitude across both outcomes. Column (2) adds region and income-group controls, after which the coefficient remains significant and of similar magnitude, ruling out the possibility that results merely reflect structural differences across country clusters. Column (3) replaces the contemporaneous spending gap with its lagged value to address reverse causality. The lagged gap remains statistically significant with the expected sign, suggesting that historical differences in education financing predict current inequality and economic performance rather than the reverse.

Across all specifications, a larger public share of education spending is associated with lower disposable income inequality and higher GDP per capita, with the direction, magnitude, and significance of the effect remaining stable throughout. This consistency across functional forms, mobility measures, initial conditions, dynamic structures, and reverse-causality tests confirms that the relationship is not an artefact of model parametrisation or omitted heterogeneity.

4 Discussion

This study shows that the composition of education finance, specifically the balance between public and private spending, plays an important role in shaping long-run growth and income inequality. Because the public–private education spending gap is expressed in levels, the coefficients in the baseline regressions can be interpreted as semi-elasticities. The inequality specification indicates that a one-unit increase in the public–private gap is associated with a 6.4 percent decrease in the disposable income Gini, corresponding to a reduction of roughly 0.019 Gini points for a country with a Gini of 0.30. The estimated semi-elasticity for GDP per capita of 0.025 implies a 2.5 percent increase per unit increase in the gap, suggesting that reallocating even modest resources toward public education can generate meaningful improvements in both equality and aggregate income. It should be noted that the empirical estimates and model counterfactuals operate on different margins, since the empirical specification measures changes in the overall spending gap across countries while the model simulations examine specific fiscal reforms, so the quantitative effects are not directly comparable but point to consistent qualitative mechanisms.

The mechanisms linking education financing to growth and inequality operate through two complementary channels. Private education spending tends to amplify inequality because it depends directly on parental income, widening skill gaps and strengthening intergenerational transmission of advantage. Public spending, by contrast, lowers the effective cost of skill investment and broadens access to human capital accumulation, compressing the skill distribution and raising the share of the population able to acquire productive skills. This broader participation in human capital formation raises aggregate productivity and supports long-run growth. These findings concern the overall balance between public and private financing rather than the ownership structure of institutions, and many successful systems combine substantial public funding with private provision. Public spending may also display diminishing returns, which helps explain why the theoretical simulations show that combining progressive taxation with increased public education spending generates the largest welfare gains.

The broader implications are consistent with evidence documented by Lindert (2004), showing that expansions of social spending have often reduced inequality without undermining growth. From a policy perspective, countries where private spending represents a large share of education finance may benefit from strengthening public subsidies, reducing private cost burdens, and ensuring that tertiary tuition policies do not make access dependent on family resources or student debt. Future research could enrich this framework by incorporating credit markets, wealth accumulation, or heterogeneous returns to education on the theoretical side, and by using microdata on household expenditures and student loans to uncover the mechanisms behind the aggregate patterns documented here.

5 Conclusion

This paper examines how the composition of education financing, specifically the balance between public and private spending, shapes long-run growth and inequality. An overlapping-generations model shows that while public and private investment are complementary, a larger public share promotes both growth and equity by low-

ering the effective cost of human capital accumulation and broadening access to skill formation. Policy simulations confirm that modest increases in public spending raise GDP per capita and reduce inequality, with the largest welfare gains arising from reforms combining progressive taxation with higher public education expenditure.

The empirical analysis corroborates these predictions. System GMM estimates on a multi-country panel show that a larger public–private spending gap is associated with significantly lower disposable income inequality and higher GDP per capita, with a one-unit increase corresponding to a 6.4 percent reduction in the Gini coefficient and a 2.5 percent increase in GDP per capita. Results are robust across alternative specifications.

These findings show that spending composition, not just spending levels, is central to distributional and growth outcomes. Policies that strengthen public investment, reduce financial barriers, or shift tuition systems toward greater cost-sharing can yield both equity and efficiency gains, particularly for countries seeking to promote social mobility and sustain long-run growth.

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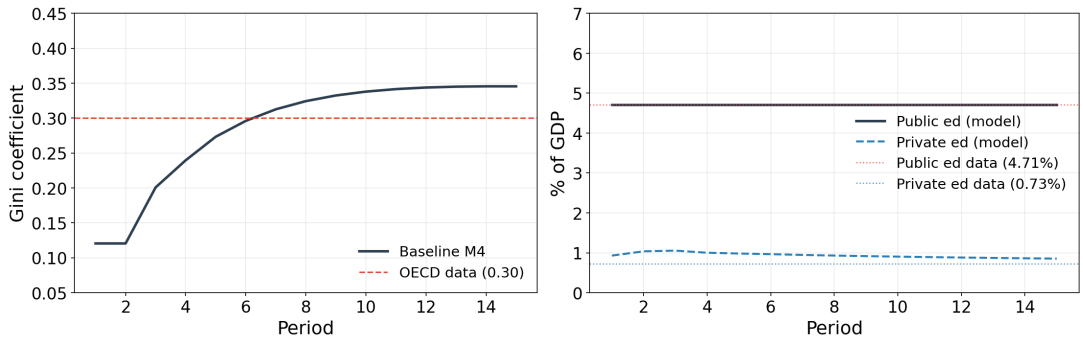
Appendix

A Theoretical Model: Additional Information

A.1 Calibration



(a) Income Inequality (b) Education Spending

Figure A.1.1: Model calibration: baseline dynamics

Note: Panel (a) shows the evolution of the Gini coefficient in the calibrated baseline economy as the model converges to its steady state. The dashed line indicates the empirical OECD benchmark (0.30). Panel (b) displays the model-implied shares of public and private education spending as a percentage of GDP during the transition, with the empirical calibration targets. Source: Author's calculations.

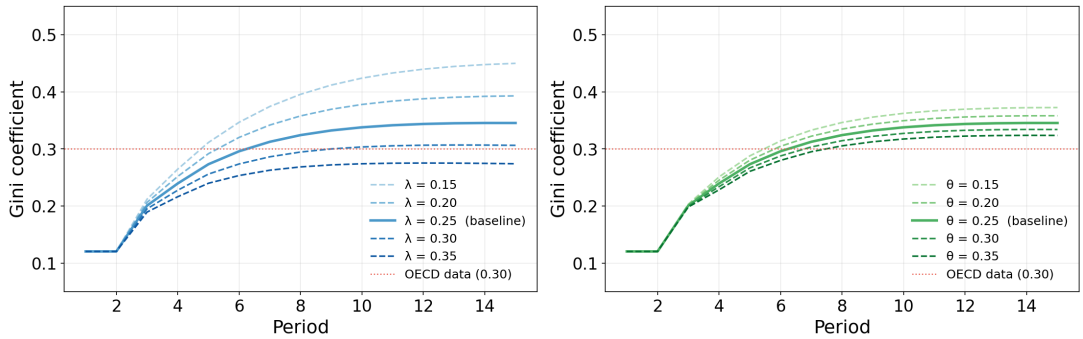
A.2 Sensitivity Analysis

This appendix reports robustness checks for alternative values of the education parameters (λ, θ). These parameters govern the relative importance of public and private educational inputs in the human capital production function. In particular, λ captures the elasticity of human capital with respect to public education spending, while θ measures the elasticity with respect to private parental investment.

Figure A.2.2 illustrates the evolution of the Gini coefficient under alternative values of these parameters. Panel (a) varies λ, the elasticity of human capital with respect to public education. When λ is lower, public spending is less effective at equalizing human capital accumulation, leading to higher long-run levels of inequality. Conversely, higher values of λ strengthen the role of public education in reducing inequality across households.

Panel (b) varies θ, the elasticity of human capital with respect to private education spending. When private educational investment becomes more productive, inequality tends to increase because higher-income households invest more in their children's education. As a result, greater reliance on private education amplifies differences in human capital accumulation across households.

Figure A.2.3 reports the corresponding dynamics for aggregate output. Higher values of λ increase the effectiveness of public education spending, strengthening human

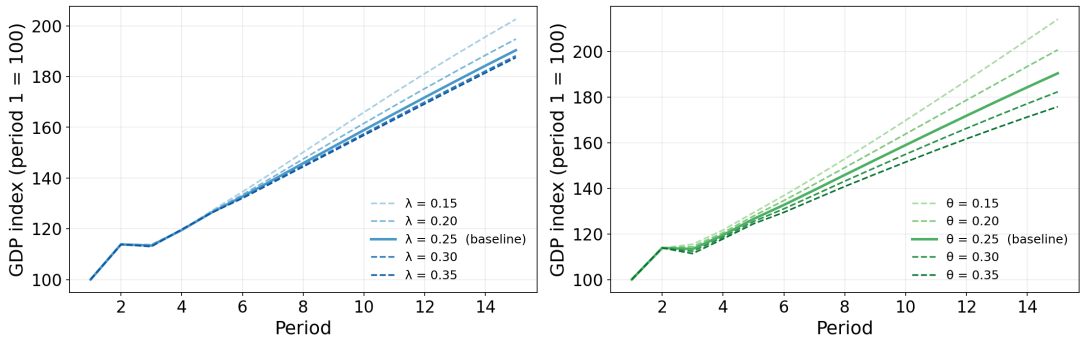


(a) Variation in λ (b) Variation in θ

Figure A.2.2: Inequality dynamics under alternative education elasticities

Notes: Evolution of the Gini coefficient under alternative values of the education elasticities. Panel (a) varies the elasticity of human capital with respect to public education (λ), while Panel (b) varies the elasticity with respect to private education (θ). The solid line corresponds to the baseline calibration ($\lambda = \theta = 0.25$). The horizontal dashed line indicates the OECD benchmark Gini coefficient (0.30). Source: Author's calculations.

capital accumulation and leading to higher GDP over time. In contrast, larger values of θ increase the role of private education spending in human capital formation. Because private investment is concentrated among higher-income households, greater reliance on private inputs tends to widen disparities in human capital accumulation and is associated with slightly lower aggregate output growth in the simulations.



(a) Variation in λ (b) Variation in θ

Figure A.2.3: GDP dynamics under alternative education elasticities

Notes: Evolution of GDP under alternative values of the education elasticities. Panel (a) varies the elasticity of human capital with respect to public education (λ), while Panel (b) varies the elasticity with respect to private education (θ). The solid line represents the baseline calibration ($\lambda = \theta = 0.25$). GDP is normalized so that period 1 equals 100. Source: Author's calculations.

Overall, the sensitivity analysis confirms that the model's qualitative predictions are robust to reasonable variation in the education elasticities. Across all parameter configurations the economy converges toward a stable growth path. Public education continues to play an important equalizing role in human capital accumulation, while stronger reliance on private education tends to amplify inequality. Although the magnitude of inequality and output dynamics varies across parameter values, the central mechanism of the model remains unchanged.

A.3 Welfare Counterfactuals

This subsection reports the welfare effects of the policy counterfactuals discussed in the main text. Welfare changes are measured using consumption-equivalent variation (CEV), which expresses the change in lifetime utility relative to the baseline steady state as the uniform percentage change in consumption that would make households indifferent between the baseline and the counterfactual equilibrium. For each household type i, welfare changes are computed from the difference in lifetime utility between the counterfactual and baseline steady states:

$$\Delta U_i = U_i^{policy} - U_i^{baseline}.$$

Given the equation 1, the consumption-equivalent welfare change is defined as

$$CEV_i = \exp\left(\frac{\Delta U_i}{1+\beta+\gamma_i}\right) - 1.$$

The denominator $(1+\beta+\gamma_i)$ reflects the marginal utility weights of the three consumption components in the household problem. Because γ_i varies across households due to non-homothetic education preferences, the scaling factor differs across individuals. Welfare comparisons are computed within the steady-state generation of households. Aggregate welfare is obtained by averaging individual welfare changes across the stationary distribution of households. Table A.3.1 reports the resulting CEV for the four policy counterfactuals considered in the model: increased public education spending (CF1), progressive taxation (CF2), the combined reform (CF3), and reduced public education spending (CF4).

Table A.3.1
Welfare Effects of Policy Counterfactuals (Steady-State CEV)

	CF1 Spending	CF2 Tax Reform	CF3 Combined	CF4 Privatise
Aggregate Welfare (CEV %)	+2.12	+10.12	+12.43	-2.11
Bottom 50% (income)	+2.29	+25.06	+27.82	-2.27
Median Household	+2.10	+5.78	+7.97	-2.09
Top 10% (income)	+1.82	-13.80	-12.19	-1.81
Change in Aggregate Welfare (ΔW)	+0.04	+0.15	+0.19	-0.04

Notes: Welfare changes are measured as consumption-equivalent variation (CEV) relative to the baseline steady state. Positive values indicate welfare gains relative to the baseline equilibrium. The CEV is computed as $CEV_i = \exp(\Delta U_i/(1+\gamma_i+\beta)) - 1$, where ΔU_i denotes the difference in lifetime utility between the policy and baseline steady states under the utility function $u(c_t, d_{t+1}, e_t) = \log(c_t) + \beta\log(d_{t+1}) + \gamma_i\log(e_t)$. Because γ_i varies across households, the scaling factor $(1+\beta+\gamma_i)$ differs across individuals. Welfare comparisons are computed within the steady-state generation of households. Source: Author's calculations.

B Empirical Analysis: Additional Information

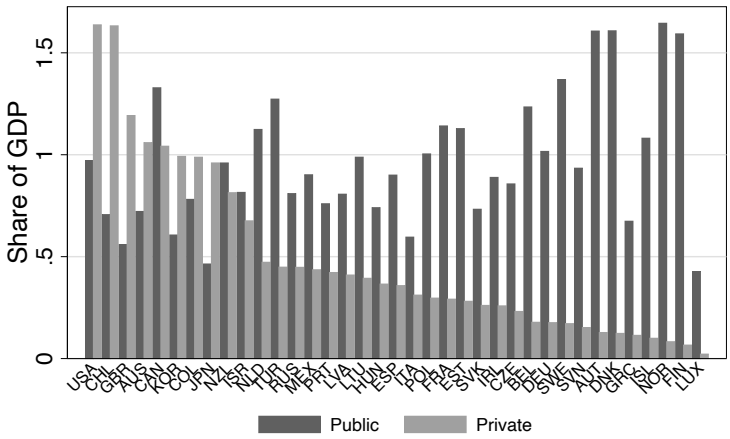


Figure B.0.4: Public and Private Spending on Tertiary Education

Note: The figure shows the shares of public and private investment in tertiary education by country. Source: OECD (2022).

Table B.0.2
List of Countries: Empirical Analysis

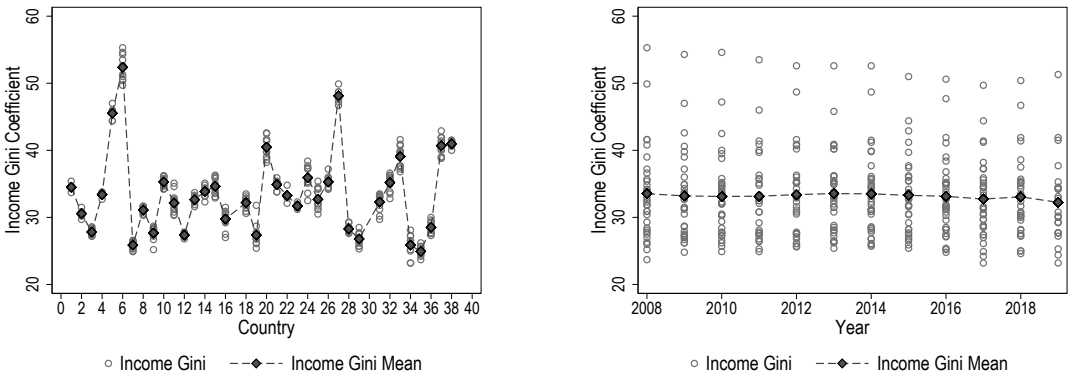
WB Code	Country	WB Code	Country	WB Code	Country
AUS	Australia	AUT	Austria	BEL	Belgium
CAN	Canada	CHL	Chile	COL	Colombia
CZE	Czech Rep.	DNK	Denmark	DEU	Germany
ESP	Spain	EST	Estonia	FIN	Finland
FRA	France	GBR	Great Britain	GRC	Greece
HUN	Hungary	IRL	Ireland	ISL	Iceland
ISR	Israel	ITA	Italy	JPN	Japan
KOR	Korea. Rep.	LTU	Lithuania	LUX	Luxembourg
LVA	Latvia	MEX	Mexico	NLD	Netherlands
NOR	Norway	NZL	New Zealand	POL	Poland
PRT	Portugal	RUS	Russian Fed.	SVK	Slovakia
SVN	Slovenia	SWE	Sweden	TUR	Turkey
USA	United States				

Note: Countries included in the empirical analysis and respective World Bank ISO codes.

Table B.0.3
Summary Statistics (37 countries from 2008-2019)

Variable	Obs.	Mean	Std.D.	Min.	Max.
Log GDP p.c.	444	10.21	0.68	8.52	11.61
Log D. Income Gini	381	3.49	0.18	3.14	4.01
Public-Private Gap	405	0.50	0.62	−1.42	1.70
Log Tertiary Enrollment	389	4.23	0.32	2.36	5.00
Governance	444	6.17	3.84	−4.72	11.20
FDI	444	−0.71	6.45	−34.86	64.51
Log GFCF	444	3.07	0.18	2.36	3.98
Log Trade	444	4.42	0.54	3.19	5.94

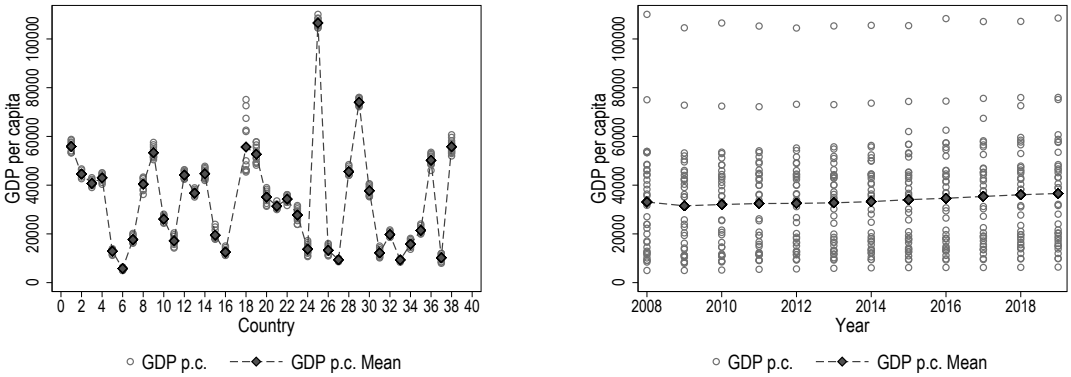
Note: This table reports summary statistics of the variables used in this analysis for 37 countries over the period 2008 - 2019. Further description of the variables in table B.0.4.



(a) Country heterogeneity (b) Time heterogeneity

Figure B.0.5: Income Inequality Heterogeneity

Note: The figure shows the Gini coefficient as a measure of income inequality. Panel (a) shows the heterogeneity of income inequality among the countries included in the main data and panel (b) across time.



(a) Country heterogeneity (b) Time heterogeneity

Figure B.0.6: GDP p.c. Heterogeneity

Note: The figure shows the heterogeneity of the sample for GDP per capita. Panel (a) shows the differences in GDP per capita among the countries included in the main data set and panel (b) across time.

Additional Instruments in Regression Analysis

Income Inequality Analysis: GDP p.h., Log of R&D, Log of market income Gini, Log of Ad. Fertility, governance, gov. consumption, Log of gov. health expenditure per capita, compulsory education, trade, inequality of opportunity of earnings, absolute and relative mobility in education, labor force with intermediate education, initial market and disposable Gini, employment in agriculture, FDI, GFCF, access to electricity, the difference between public and private expenditure on health, total enrollment tertiary education, tax revenue, hospital beds per 1000 people, CO2 emissions, taxes, and inequality of opportunity of earnings.

GDP per capita Analysis: Access to electricity, hospital beds per 1000 people, population growth rate, labor force with an intermediate level of education, adolescent fertility, gov. expenditure health, labor productivity, R&D expenditure, the difference between public and private expenditure on health, compulsory years of education, taxes (different) the measure of inequality of opportunity and the income share bottom 50%, fertility rate, labor force, Ad. Fertility.

Note: All the additional instruments listed for each dependent variable are included in the main analysis and the robustness checks, but are not necessarily used in every dynamic panel data regression.

Table B.0.4
Definition and Sources of Variables

Variable	Description and Source
Public-Private Gap	Total public minus private expenditure on the highest level of education. Public and private spending measures are a percentage of GDP. Source: OECD (2022).
Disposable Income Gini	Gini index measures the extent of the equality of the distribution. Estimates of Gini index of 0 represent perfect equality, while an index of 100 implies perfect inequality. Source: World Bank, Development Research Group.
GDP p.c.	GDP per capita (constant 2015 US$). Source: "World development indicators"
Tertiary Enrollment	Ratio of total enrollment to the population of the age group that corresponds to the level of education. Source: UNESCO Institute for Statistics.
Governance	The sum of control of corruption, government effectiveness, political stability and absence of violence, voice and accountability, regulatory quality, and rule of law. Indicators range from -2.5 to 2.5. Source: The World Governance Indicators.
FDI	Annual net foreign direct investment (share of GDP). Source: National Accounts Data, World Bank.
GFCF	Annual gross fixed capital formation (share of GDP). Source: National Accounts Data, World Bank.
Trade	Annual trade (share of GDP). Source: National Accounts Data, World Bank.
Inflation	Annual inflation measured by the consumer price index. Source: International Financial Statistics, International Monetary Fund.
Education	General government expenditure on education (share of GDP). Source: UNESCO Institute for Statistics.
Top 1% (10%)	Pre-tax national income share held by the top 1% (10%) of the income distribution. Source: World Inequality Database.
Bottom 50%	Pre-tax national income share held by the bottom 50% of the income distribution. Source: World Inequality Database.
GDP p.h.	GDP per total hours worked of all people engaged in production. This indicator is measured in USD. Source: OECD Data.
Absolute Mobility	The measure for absolute mobility is the share of respondents that have attained a higher educational category than their parents, conditional on the parents not having obtained tertiary education. Source Van der Weide et al. (2021).
Relative Mobility	The measure for relative mobility is one minus the correlation coefficient from the regression of children's years of education on the education of their parents. Source Van der Weide et al. (2021).
Public Ex. Ed.	Includes direct expenditure on educational institutions, educational-related public subsidies given to households and administered by educational institutions.
Private Ex. Ed.	Includes all direct expenditure on educational institutions, net of public subsidies, excluding expenditure outside educational institutions such as textbooks purchased by families, private tutoring for students and student living costs.

Note: Variables included in the empirical analysis with their respective description and sources.

C Empirical Analysis: Robustness Tests

Table C.0.5
Dynamic Regression: Income Inequality — Technical Robustness

	Log Disposable Income Gini		
	(1)	(2)	(3)
L.Log D. Income Gini	0.926***	0.875***	0.957***
	(0.044)	(0.058)	(0.049)
Public-Private Gap	−0.047**	−0.063***	
	(0.020)	(0.022)	
L.Public-Private Gap			−0.049**
			(0.018)
Observations	226	226	223
Hansen p-value	0.125	0.632	0.232
AR(2) p-value	0.855	0.734	0.726
No. Instruments	20	21	20
No. Countries	23	23	23

Note: This table presents robustness for the log disposable income Gini baseline. Column (1) adds dynamic extensions, (2) includes region–income group controls, and (3) tests reverse causality using lagged public–private education spending. Models use one-step System GMM with collapsed instruments and orthogonal deviations. Standard errors are Windmeijer-corrected and robust to heteroscedasticity. Constant, country fixed effects, and controls are included but omitted for brevity. Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table C.0.6
Dynamic Regression: GDP per Capita — Technical Robustness

	Log GDP per Capita		
	(1)	(2)	(3)
L.Log GDP p.c.	0.983***	0.978***	0.978***
	(0.012)	(0.022)	(0.020)
Public-Private Gap	0.027*	0.039**	
	(0.014)	(0.016)	
L.Public-Private Gap			0.038***
			(0.013)
Observations	233	233	227
Hansen p-value	0.317	0.160	0.131
AR(2) p-value	0.251	0.458	0.672
No. Instruments	27	27	26
No. Countries	32	32	32

Note: This table presents robustness for the log GDP per capita baseline. Column (1) adds dynamic extensions, (2) includes region–income group controls, and (3) tests reverse causality using lagged public–private education spending. Models use one-step System GMM with collapsed instruments and orthogonal deviations. Standard errors are Windmeijer-corrected and robust to heteroscedasticity. Constant, country fixed effects, and controls are included but omitted for brevity. Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table C.0.7
Dynamic Regression: Income Inequality — Education Robustness

	Log Disposable Income Gini		
	(1)	(2)	(3)
L.Log D. Income Gini	0.911***	0.835***	0.807***
	(0.062)	(0.136)	(0.099)
Public-Private Gap Sq.	0.027**		
	(0.011)		
Public+Private E.E.		0.040	
		(0.046)	
Public/Private Gap			−0.005**
			(0.002)
Public-Private Gap	−0.079***	−0.084**	
	(0.028)	(0.039)	
Observations	226	226	263
Hansen p-value	0.309	0.245	0.715
AR(2) p-value	0.894	0.791	0.367
No. Instruments	21	20	21
No. Countries	23	23	27

Note: This table presents robustness tests of the effect of education variables on the log disposable income Gini. Column (1) includes the squared public–private education gap. 2) includes total public+private education expenditure. (3) tests the ratio of public to private education spending. Standard errors are Windmeijer-corrected and are robust to heteroscedasticity. Country fixed effects and constants included but omitted for brevity. Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table C.0.8
Dynamic Regression: GDP per Capita — Education Robustness

	Log GDP per Capita		
	(1)	(2)	(3)
L.Log GDP p.c.	0.964***	0.940***	0.979***
	(0.025)	(0.035)	(0.018)
Public-Private Gap Sq.	0.016		
	(0.014)		
Public+Private E.E.		−0.045	
		(0.044)	
Public/Private Gap			0.004**
			(0.002)
Public-Private Gap	0.064**	0.057**	
	(0.029)	(0.026)	
Observations	233	242	233
Hansen p-value	0.097	0.260	0.179
AR(2) p-value	0.209	0.328	0.578
No. Instruments	25	26	27
No. Countries	32	32	32

Note: This table presents robustness tests of the effect of education variables on the log GDP per capita. Column (1) includes the squared public–private education gap. 2) includes total public+private education expenditure. (3) tests the ratio of public to private education spending. Standard errors are Windmeijer-corrected and are robust to heteroscedasticity. Country fixed effects and constants are included but omitted for brevity. Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table C.0.9
Dynamic Regression: Income Inequality — Mobility and Interaction Robustness

	Log Disposable Income Gini			
	(1)	(2)	(3)	(4)
L.Log D. Income Gini	0.933***	0.919***	0.975***	0.933***
	(0.057)	(0.057)	(0.164)	(0.048)
Abs. Mobility	−0.021			
	(0.046)			
Rel. Mobility		0.010		
		(0.090)		
D.Gini × I.M.Gini			−0.006	
			(0.043)	
GDP p.c. × I.GDP p.c.				0.000
				(0.000)
Public–Private Gap	−0.064**	−0.065***	−0.047*	−0.055**
	(0.025)	(0.023)	(0.026)	(0.022)
Observations	226	226	226	221
Hansen p-value	0.310	0.294	0.405	0.497
AR(2) p-value	0.846	0.834	0.898	0.776
No. Instruments	21	21	21	21
No. Countries	23	23	23	22

Note: This table presents four robustness specifications of the System GMM model for log Income Gini. This table reports robustness checks using alternative mobility and interaction specifications. Standard errors are Windmeijer-corrected and robust to heteroscedasticity. Country fixed effects, constants, and control variables are included but omitted for brevity. Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table C.0.10
Dynamic Regression: GDP per Capita — Mobility and Interaction Robustness

	Log GDP per Capita			
	(1)	(2)	(3)	(4)
L.Log GDP p.c.	0.951***	1.010***	0.934***	0.990***
	(0.028)	(0.054)	(0.041)	(0.013)
Abs. Mobility	0.079			
	(0.084)			
Rel. Mobility		−0.146		
		(0.242)		
GDP p.c. x I. GDP p.c.			0.000	
			(0.000)	
D. Gini x I.M. Gini				0.001
				(0.007)
Public-Private Gap	0.069*	0.066**	0.117***	0.025**
	(0.037)	(0.026)	(0.041)	(0.011)
Observations	189	197	242	233
Hansen p-value	0.221	0.070	0.258	0.075
AR(2) p-value	0.226	0.907	0.197	0.155
No. Instruments	26	25	24	28
No. Countries	26	26	32	32

Note: This table presents four robustness specifications of the System GMM model for log GDP per capita. This table reports robustness checks using alternative mobility and interaction specifications. Standard errors are Windmeijer-corrected and robust to heteroscedasticity. Country fixed effects, constants, and control variables are included but omitted for brevity. Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.